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United States Patent	12404960
Kind Code	B1
Date of Patent	September 02, 2025
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Pipe connector apparatus

Abstract

A connector apparatus featuring a main pipe with first and second opposed ends. Each has a corresponding male-threaded fastening section for mating with the complimentary female-threaded fastening section on a respective conventional pipe-locking ring or collar. A valve-housing for supporting a one-way flow valve assembly interiorly, and a riser mounted on and supported by the valve housing for ultimate connection to the output or exhaust nozzle of a medical scavenger device.

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Appl. No.:	18/763582
Filed:	July 03, 2024

Related U.S. Application Data

continuation parent-doc US 18336402 20230616 US 12049973 child-doc US 18763582
continuation parent-doc US 17651710 20220218 US 11732827 20230822 child-doc US 18336402
continuation-in-part parent-doc US 17248138 20210111 US 11313113 20220426 child-doc US 17651710
us-provisional-application US 62995133 20200114

Publication Classification

Int. Cl.: F16L37/00 (20060101); E03C1/122 (20060101)

U.S. Cl.:

CPC **F16L37/008** (20130101); **E03C1/122** (20130101);

Field of Classification Search

CPC: F16L (37/008); E03C (1/122)

USPC: 137/343; 137/247.41; 137/362; 137/577; 4/679; 4/378; 4/584; 4/613; 4/650; 4/668; 4/671;
25/134.1

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This nonprovisional application is a continuation of and claims priority to U.S. Nonprovisional patent application Ser. No. 18/336,402, entitled “PIPE CONNECTOR APPARATUS,” filed Jun. 16, 2023 by the same inventor, which is a continuation of and claims priority to U.S. Nonprovisional patent application Ser. No. 17/651,710, entitled “PIPE CONNECTOR APPARATUS,” filed Feb. 18, 2022 by the same inventor, issued Aug. 22, 2023 as U.S. Pat. No. 11,732,827, which is a continuation-in-part of and claims priority to U.S. Nonprovisional patent application Ser. No. 17/248,138, entitled “PIPE CONNECTOR APPARATUS,” filed Jan. 11, 2021 by the same inventor, issued Apr. 26, 2022 as U.S. Pat. No. 11,313,113, and also claims priority to U.S. Provisional Patent Application Ser. No. 62/995,133, entitled “PIPE CONNECTOR APPARATUS,” filed Jan. 14, 2020 by the same inventor.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) This invention relates, generally, to pipe connector assemblies. More specifically, it relates to a pipe connector apparatus or assembly for connecting the exhaust line of a medical gas scavenging device to a drainpipe in a building or other structure for discharge into the building's wastewater collection system.

2. Brief Description of the Prior Art

(2) It is frequently necessary to vent the exhaust gas emanating from a medical device, such as a gas analgesia machine, outside of the immediate location or treatment room where the medical device is being used. This is currently accomplished by placing a pipe vent in a wall of the treatment room to exhaust the waste gas outside the corresponding building structure. However, when the medical device producing the waste gas is located in an interior room, with no exterior building walls, such a solution becomes impracticable.

(3) To overcome this problem, it is necessary to employ an efficient venting system to remove medical waste gas from interior rooms. Such a venting system includes a connection between the exhaust hose of a medical gas scavenging device to a plumbing fixture drainpipe within a room, such as a sink or washbasin as is common in medical office or building.

(4) Accordingly, what is needed is an in-line pipe connector apparatus capable of discharging an exhalation gas safely and effectively from a treatment room. What is also needed is an in-line pipe connector apparatus that is adaptable to be installed under a conventional washbasin downstream from the trap. However, in view of the art considered as a whole at the time the present invention was made, it was not obvious to those of ordinary skill in the field of this invention how the shortcomings of the prior art could be overcome.

(5) All referenced publications are incorporated herein by reference in their entirety. Furthermore, where a definition or use of a term in a reference, which is incorporated by reference herein, is inconsistent or contrary to the definition of that term provided herein, the definition of that term provided herein applies and the definition of that term in the reference does not apply.

(6) While certain aspects of conventional technologies have been discussed to facilitate disclosure of the invention, Applicant in no way disclaims these technical aspects, and it is contemplated that the claimed invention may encompass one or more of the conventional technical aspects discussed herein.

(7) The present invention may address one or more of the problems and deficiencies of the prior art discussed above. However, it is contemplated that the invention may prove useful in addressing other problems and deficiencies in a number of technical areas. Therefore, the claimed invention should not necessarily be construed as limited to addressing any of the particular problems or deficiencies discussed herein.

(8) In this specification, where a document, act, or item of knowledge is referred to or discussed, this reference or discussion is not an admission that the document, act or item of knowledge or any combination thereof was at the priority date, publicly available, known to the public, part of common general knowledge, or otherwise constitutes prior art under the applicable statutory provisions; or is known to be relevant to an attempt to solve any problem with which this specification is concerned.

BRIEF SUMMARY OF THE INVENTION

(9) An aspect of the present disclosure provides a pipe-connector assembly system for exhausting a waste gas from an interior environment of a building. The pipe-connector assembly system comprises a pipe connector apparatus, which includes, a main pipe member extending between a first member end and a second member end, a valve housing including a proximal end, a distal end, and a valve residing within the valve housing and disposed between the proximal end and the distal end of the valve housing. In embodiments, the proximal end is connected with the main pipe member and the distal end is connected with a gas scavenger device. The system also comprises a drainpipe having a first end connected with the second member end of the pipe connector apparatus and a second end connected with a waste collection system of the building. In some embodiments, the waste gas flows from the gas scavenger device through the pipe connector apparatus and into the waste collection system of the building, such that the waste gas is safely evacuated from the interior environment.

(10) In embodiments, the pipe-connector assembly system also comprises a quick-release valve having an inlet, an output, and a passage extending between the inlet and the output. The inlet is configured to be removably coupled to an exhaust hose of the gas scavenger device, while the output is connected with the distal end of the valve housing. The waste gas travels from the gas scavenger device through the quick-release valve to the pipe connector apparatus and is exhausted into the waste-collection system of the building.

(11) In embodiments, a riser is disposed between the distal end of the valve housing and the gas scavenger device. The riser includes a body defining a bore extending between a first riser end coupled to the distal end of the valve housing and a second riser end coupled to an exhaust hose of the gas scavenger device. The riser is fluidly connected with the main pipe member and the gas scavenger device.

(12) In embodiments, pipe-connector assembly system also comprises a plumbing trap. The plumbing trap includes a first trap end and a second trap end. The first trap end is configured to be connected with a washbasin disposed within the interior environment. Accordingly, a first attachment mechanism is disposed at the first member end and configured to threadedly engage with a second attachment mechanism disposed at the second trap end of the plumbing trap. In addition, a third attachment mechanism is disposed at the second member end and configured to threadedly engage with the first end of the drainpipe, such that the pipe connector apparatus is

secured between the plumbing trap and the drainpipe.

(13) In embodiments, the valve is a one-way valve, such that that the backflow of the waste gas from the waste collection system back into the internal environment is prevented. In embodiments, the valve further includes a closed position and an open position, where the open position permits the waste gas to flow through the valve housing from the internal environment to the waste collection system of the building. In the closed position, the valve prevents the backflow of the waste gas from the waste collection system back into the internal environment. In embodiments the valve comprises of a biasing component that applies a biasing force on a rigid portion towards the closed position. The rigid portion is translatable between the closed position and the open position when an external force, emanating from the gas scavenger device, overcomes the biasing force. In embodiments, the waste gas is a medical waste gas generated by an analgesic system fluidly connected with the gas scavenger system. In some embodiments, the waste gas comprises a mixture of nitrous oxide and oxygen.

(14) In another aspect, the present disclosure provides a pipe-connector assembly system for exhausting a medical waste gas from a treatment room within a building. The pipe-connector assembly system comprises a pipe connector apparatus, which includes, a main pipe member extending between a first member end and a second member end, the first member end connected with the second trap end, a valve housing including a proximal end, a distal end, and a one-way valve residing within the valve housing and disposed between the proximal end and the distal end of the valve housing. The proximal end is connected with the main pipe member and the distal end is connected with a medical gas scavenger device. The one-way valve is configured to be translatable between an open position, which is configured to permit the flow of the medical waste gas through the valve housing, and a closed position, which is configured to prevent the flow of the medical waste gas through the valve housing. The one-way valve is biased towards the closed position. The pipe-connector assembly system also comprises a drainpipe having a first end connected with the second member end of the pipe connector apparatus and a second end connected with a waste collection system of the building. In some embodiments, the medical waste gas flows from the medical gas scavenger device through the pipe connector apparatus and into the waste collection system of the building, such that the medical waste gas is safely evacuated from the treatment room.

(15) In embodiments, the pipe-connector assembly system includes a quick-release valve which has an inlet, an output, and a passage that extends between the inlet and the output. The inlet is configured to be removably coupled to an exhaust hose of the medical gas scavenger device, and the output is connected with the distal end of the valve housing, allowing the medical waste gas to travel from the medical gas scavenger device through the quick-release valve to the pipe connector apparatus and exhausted into the waste-collection system of the building.

(16) In embodiments, a riser is disposed between the distal end of the valve housing and the medical gas scavenger device. The riser includes a body defining a bore extending between a first riser end coupled to the distal end of the valve housing and a second riser end coupled to an exhaust hose of the medical gas scavenger device. The riser is fluidly connected with the main pipe member and the medical gas scavenger device.

(17) In embodiments, pipe-connector assembly system also comprises a plumbing trap. The plumbing trap includes a first trap end and a second trap end. The first trap end is configured to be connected with a washbasin disposed within the interior environment. Accordingly, a first attachment mechanism is disposed at the first member end and configured to threadedly engage with a second attachment mechanism disposed at the second trap end of the plumbing trap. In addition, a third attachment mechanism is disposed at the second member end and configured to threadedly engage with the first end of the drainpipe, such that the pipe connector apparatus is secured between the plumbing trap and the drainpipe.

(18) In embodiments, the medical waste gas comprises a mixture of nitrous oxide and oxygen. In

embodiments, a biasing component applies a biasing force on a rigid portion towards the closed position, preventing the flow of the medical waste gas within the valve assembly. The rigid portion is translatable between the closed position and the open position when an external force, emanating from the gas scavenger device, overcomes the biasing force.

(19) In embodiments, the valve housing includes an interior wall, extending between the proximal end and the distal end. A channel is defined by the interior wall. The valve housing also includes a support floor member disposed within the channel. The support floor member comprises a central hub, which has a central valve-step support aperture and a rib extending radially between the central hub and the interior wall of the channel.

(20) In another aspect, the present disclosure provides a method of exhausting a medical waste gas scavenged by a medical gas scavenging device from a treatment room in a building. The method comprises (a) coupling a second member end of a pipe connector apparatus to a drainpipe, the drainpipe fluidly connected with a waste collection system of the building; (b) coupling an exhaust hose fluidly connected with the medical gas scavenging device to the distal end of the valve housing; (c) collecting, via the medical gas scavenging device, an amount of the medical waste gas emitted; (d) transferring, via the exhaust hose, an amount of the medical waste gas emitted by the medical gas scavenging device into the pipe connector; and (e) exhausting, via the drainpipe, the amount of the medical waste gas into the waste collection system of the building.

(21) In embodiments, the pipe connector apparatus includes a second member end opposite the first member end and a main pipe member extending therebetween, a valve housing including a proximal end opposite a distal end, and a valve residing within the valve housing and disposed between the proximal end and the distal end. The valve housing extends orthogonally away from the main pipe member between the first member end and the second member end.

(22) In embodiments, the steps of the method of the present disclosure include providing a quick release valve, where the quick release valve has an inlet, an output, and a passage extending between the inlet and the output. The method also includes coupling the inlet to the exhaust hose of the medical gas scavenger device; coupling the outlet to the distal end of the valve housing; and transferring, via the quick-release valve, the medical waste gas from the medical gas scavenging device to the pipe connector apparatus.

(23) In embodiments, the steps of the method also include identifying the drainpipe disposed between the medical scavenging device and waste collection system of the building; placing a cutting device onto the drainpipe, where the cutting device is configured to cut the drainpipe into a first section and a second section; operating the cutting device, such that the cutting device cuts the drainpipe into the first section and the second section; and installing the pipe connector apparatus between the first section and the second section, such that the first member end of the pipe connector is connected with the first section, and such that the second member end of the pipe connector apparatus is connected with the second section. The pipe connector apparatus is connected with the medical scavenging device and the waste collection system of the building.

(24) These and other important objects, advantages, and features of the invention will become clear as this disclosure proceeds.

(25) The invention accordingly comprises the features of construction, combination of elements, and arrangement of parts that will be exemplified in the disclosure set forth hereinafter, and the scope of the invention will be indicated in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a fuller understanding of the invention, reference should be made to the following detailed description, taken in connection with the accompanying drawings, in which:

- (2) FIG. 1 is a schematic view of an embodiment of the pipe connector apparatus of the present invention in fluid communication with an analgesia system according to embodiments of the present disclosure.
- (3) FIG. 2 is a perspective view of an embodiment of the pipe connector apparatus of the present invention according to embodiments of the present disclosure.
- (4) FIG. 3 is a side view of an embodiment of the pipe connector apparatus of FIG. 2 according to embodiments of the present disclosure.
- (5) FIG. 4 is a side view of an embodiment of the pipe connector apparatus of FIG. 2 according to embodiments of the present disclosure.
- (6) FIG. 5 is a top plan view of an embodiment of the pipe connector apparatus of FIG. 2 according to embodiments of the present disclosure.
- (7) FIG. 6 is a cross-sectional view of an embodiment of the pipe connector apparatus of FIG. 5 taken along line C-C according to embodiments of the present disclosure.
- (8) FIG. 7 is a cross-sectional view Detail E of FIG. 6 according to embodiments of the present disclosure.
- (9) FIG. 8 is an enlarged view of the pipe connector apparatus of FIG. 7, particularly showing the circled portion D, depicting the valve in an open position according to embodiments of the present disclosure.
- (10) FIG. 9 is a top cross-sectional view of an embodiment of the pipe connector apparatus taken along line B-B of FIG. 4 according to embodiments of the present disclosure.
- (11) FIG. 10 is an exploded assembly view of an embodiment of the pipe connector apparatus of the present invention in association with a conventional trap and drainpipe set up according to embodiments of the present disclosure.
- (12) FIG. 11 is an exploded assembly view of an embodiment of the pipe connector apparatus of the present invention in association with an embodiment of a conventional trap and drainpipe set up according to embodiments of the present disclosure.
- (13) FIG. 12 is a perspective view of an embodiment of the pipe connector apparatus of the present invention according to embodiments of the present disclosure.
- (14) FIG. 13 is a perspective view of an embodiment of the pipe connector apparatus of the present invention according to embodiments of the present disclosure.
- (15) FIG. 14 is a cross-sectional view of an embodiment of the pipe connector apparatus of FIG. 13 along line A-A according to embodiments of the present disclosure.
- (16) FIG. 15 is a top cross-sectional view of an embodiment of the pipe connector apparatus of FIG. 14 according to embodiments of the present disclosure.
- (17) FIG. 16 is an exploded assembly view of an embodiment of the pipe connector apparatus of the present invention according to embodiments of the present disclosure.
- (18) FIG. 17 is a flow chart depicting the steps of a method of evacuating a medical waste gas from a treatment room within a building.

DETAILED DESCRIPTION OF THE INVENTION

- (19) In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings, which form a part thereof, and within which are shown by way of illustration specific embodiments by which the invention may be practiced. It is to be understood that other embodiments may be utilized, and structural changes may be made without departing from the scope of the invention.
- (20) Elements/components shown in diagrams are illustrative of exemplary embodiments of the disclosure and are meant to avoid obscuring the disclosure. Reference in the specification to “one embodiment,” “preferred embodiment,” “an embodiment,” or “embodiments” means that a particular feature, structure, characteristic, or function described in connection with the embodiment is included in at least one embodiment of the disclosure and may be in more than one embodiment. The appearances of the phrases “in one embodiment,” “in an embodiment,” “in

embodiments,” “in other embodiments” or “in some embodiments” in various places in the specification are not necessarily all referring to the same embodiment or embodiments. The terms “include,” “including,” “comprise,” and “comprising” shall be understood to be open terms and any lists that follow are examples and not meant to be limited to the listed items.

(21) Any headings used herein are for organizational purposes only and shall not be used to limit the scope of the description or the claims. Furthermore, the use of certain terms in various places in the specification of for illustration and should not be construed as limiting.

(22) As used in this specification and the appended claims, whenever the term “at least,” “greater than,” or “greater than or equal to” precedes the first numerical value in a series of two or more numerical values, the term “at least,” “greater than,” or “greater than or equal to” applies to each of the numerical values in that series of numerical values. For example, greater than or equal to 1, 2, or 3 is equivalent to greater than or equal to 1, greater than or equal to 2, or greater than or equal to 3.

(23) As used in this specification and the appended claims, whenever the term “no more than,” “less than” or “less than or equal to” precedes the first numerical value in a series of two or more numerical values, the term “no more than,” “less than,” or “less than or equal to” applies to each of the numerical values in that series of numerical values. For example, less than or equal to 3, 2, or 1 is equivalent to less than or equal to 3, less than or equal to 2, or less than or equal to 1.

(24) As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the context clearly dictates otherwise.

(25) The present invention includes a simple, compact, and inexpensive connector assembly which easily may be installed in a drainpipe of a washbasin without impairing the operation of the plumbing trap, such as a P-trap, within the corresponding washbasin. The unitary in-line pipe connector assembly (alternatively referred to as “pipe connector apparatus”) includes a main pipe member having first and second opposed ends. Each end includes a corresponding male-threaded fastening section for mating with a complementary female-threaded fastening section on a respective conventional pipe-locking ring or collar. In addition, each end includes a valve-housing for supporting an interior one-way flow valve assembly, as well as a riser mounted on and supported by the valve housing for ultimate connection to the output or exhaust nozzle of a medical scavenger or evacuation device.

(26) In use, a section of the drainpipe proximal to output side of the plumbing trap is removed, and the connector apparatus of the invention is lockingly sealed into place in the drainpipe in place of the removed section. The scavenger exhaust nozzle is then connected to the riser on the connector apparatus valve housing. Exhaust gas from the scavenger flows through the one-way valve into the building waste drainage system. The unitary in-line pipe connector assembly prevents the back-flowing of the gas from the waste system by the blocking action of the one-way valve.

(27) Turning initially to FIGS. **1-17**, unitary in-line pipe connector assembly **10** (hereinafter “pipe connector apparatus **10**”) comprises three primary components: main pipe member **12**, valve housing **14**, and riser **16** extending upwardly away from valve housing portion **14**.

(28) Main Pipe Member

(29) Main pipe member **12** is cylindrically shaped, defining a hollow bore or central passage **18** extending between first peripheral end edge **20** and second peripheral edge **22**, such that central passage **18** is formed throughout main pipe member **12**. As such, first peripheral end edge **20** defines first opening **21**, and second peripheral end edge **22** defines second opening **23**, such that central passage **18** is formed throughout main pipe member **12**.

(30) Additionally, main pipe member **12** includes continuous interior wall surface **24** that may be substantially cylindrical and smooth, such that interior wall surface **24** is free from projections or unevenness. In contrast, exterior surface **26** of main pipe member **12** includes first and second

spiral fastening sections **28, 30**, which may be male-threaded fastening sections, as shown in FIG. **3**. As will be described in greater detail below, first and second spiral male-threaded fastening sections **28, 30** mechanically couple with a corresponding complementary spiral female-threaded sections of conventional drainpipe locking rings or collars. In an embodiment, fastening sections **28, 30** may be any fastening means, such that main pipe member **12** is mechanically coupled with a corresponding complementary fastening section of conventional drainpipe locking rings or collars.

(31) Referring to FIGS. **2-9**, main pipe member **12** includes valve-housing mounting base **32** (alternatively referred to as “support boss **32**”) extending orthogonally with respect to central longitudinal axis **13** of main pipe member **12**. In an embodiment, support boss **32** is monolithically formed with main pipe member **12**. In yet another embodiment, support boss **32** is formed of a separate piece, such that support boss **32** is coupled with main pipe member **12** during the installation or assembly of pipe connector apparatus **10** (e.g., such as when pipe connector apparatus **10** is installed with an existing waste-collection system of a building).

(32) Support boss **32** includes support boss body **33** extending from first boss end **35** to second boss end **39**. Formed within a portion of support boss body **33**, peripheral support end edge **34** resides between first boss end **35** and second boss end **39**. Peripheral support edge **34** is configured to receive lower portion **47** of valve housing **14**, which will be discussed in greater detail below.

(33) Additionally, support boss **32** includes passageway **37** in fluid communication with central passage **18** of main pipe member **12**. Passageway **37** includes first passage **38** and second passage **40**. First passage **38** is defined by first interior wall **43**, and second passage **40** is defined by second interior wall **41**. First interior wall **43** has a greater width (or diameter) than second interior wall **41**. In an embodiment, first interior wall **43** has a width (or diameter) equal to or less than second interior wall **41**.

(34) In an embodiment, annular abutment **42** of second boss end **39** extends at least partially within central passage **18** of main pipe member **12**. The partial protrusion of annular abutment **42** into central passage **18** serves as an axial stop member to limit the insertion of end edges **96, 98** (see FIG. **10**) longitudinally into central passage **18** when installed with the waste system of the building. In other words, annular abutment **42** prevents one or both of end edges **96, 98** from being disposed within central passage **18** in such a manner that the flow of the fluid from riser **16** to central-passage **18** is restricted or prevented entirely. In an embodiment, the fluid is a medical waste gas, an exhalation gas, waste gas, sewer gas, or any other undesired gas or liquid that a user wishes to exhaust from an environment with the user. In such an embodiment, the environment may be a medical treatment room, storage room, laboratory, or any other confined space that a user seeks to exhaust the fluid from.

(35) Riser

(36) As depicted in FIGS. **6** and **7**, riser **16** is preferably comprised of a hollow tubular structure having top-end edge **66** and opposed bottom end edge **68** defining central passage **70** extending therethrough. Bottom end edge **68** is configured to be disposed adjacent to top outer circumferential surface **49** of riser-support flow member **50** of valve-housing **14**. Furthermore, top-end edge **66** of riser **16** is in mechanical communication with the environment of the building where the fluid is being exhausted (or removed) from. In an embodiment, top-end edge **66** is mechanically coupled to a medical gas scavenging device's exhaust nozzle. In such embodiments, the medical gas scavenging device facilitates the flow of the fluid from the environment, through pipe connector apparatus **10**, and ultimately into the building's waste-collection system.

(37) Valve Housing

(38) Referring now to FIGS. **6-9**, valve-housing **14** controls the fluid flow between riser **16** and passageway **37** via valve assembly **72** disposed within cavity **64**. Specifically, valve-housing **14** prevents the fluid's backflow through pipe connector apparatus **10**, through riser **16**, and ultimately back into the environment with the user.

(39) Valve-housing **14** includes upper portion **44**, central portion **45**, and lower portion **47**. Upper

portion **44** of valve-housing **14** is preferably tapered and terminates distally, forming receipt **46** for receiving bottom portion **15** of riser **16**. Receipt **46** is defined by first internal wall **51** extending from top peripheral end edge **48** of tapered valve-housing upper portion **44** downwardly to riser-support floor member—generally denoted as reference numeral **50**. Riser floor member **50** abuts a portion of second internal wall **53** of central portion **45** of valve-housing portion **14**, thereby forming cavity **64** extending between riser floor member **50** and first boss end **35** of support boss **32**. Furthermore, riser floor member **50** includes central hub **52** defining central valve-step support opening **54**. At least one rib **56** extends radially between hub **52** and upper portion **44** of valve-housing forming passage **58**. In an embodiment, three equidistantly spaced-apart ribs **56** extend radially between hub **52** and upper portion **44** of valve-housing **14**, thereby forming passages **58A**, **58B**, and **58C** (collectively referred to as passages **58**). Lower portion **47** of valve-housing **14** is slidably disposed over and positioned adjacent to first boss end **35**. In such configurations, peripheral support edge **34** abuts terminal edge **71** support boss **32**.

(40) As noted above, valve assembly **72** regulates the fluid flow between riser **16** and central passage **18** of main pipe member **12**. The regulation of the fluid flow may be achieved by transitioning flexible (or umbrella) portion **74** between a closed position (see FIG. **6**) and an open position (see FIG. **7**, which shows circled portion D of FIG. **6**). In particular, flexible portion **74** is integrally mounted on supporting spindle **76**. Fastening enlargement or nub **78** is provided on spindle **76** and is spaced apart from flexible portion **74**. Nub **78** is sized to enable spindle **76** to be press-fit into and through central support opening **54** in floor member **50**. By doing so, valve assembly **72** is locked in place and supported by floor member **50** in an operational position (e.g., open or closed position), as shown in FIGS. **6** and **7**.

(41) In an embodiment, valve assembly **72** is biased toward the closed position, such that flexible portion **74** is concave (see FIG. **6**). Accordingly, when valve assembly **72** is in the closed position, the concave nature of flexible portion **74** completely seals off passages **58** from being in fluid communication with cavity **64**. Thus, the backflow of any fluid back into riser **16** is prevented.

(42) Alternatively, when flexible portion **74** is acted upon by the force of a fluid (such as flowing gas from a medical gas scavenging device) traveling through central passage **70** of riser **16**, flexible portion **74** flexes and assumes the open position having a convex shape (see FIG. **7**). In such open positions, any gas or other fluid flowing through riser **16** will continue to flow as indicated by arrows **80** through passages **58**. The fluid then flows into cavity **64** and through passageway **37**. Lastly, the fluid exits into central passage **18** of main pipe member **12**, where the fluid can be safely discharged into the building's waste-collection system.

(43) Thus, by providing valve assembly **72** in accordance with the present invention, the pipe connector apparatus **10** of FIGS. **2-9** is capable only of uni-directional exhaust flow. Accordingly, any backflow of an undesirable fluid (e.g., sewer gas) into the treatment room or area that may emanate from the building's waste-collection system to which pipe connector apparatus **10** is connected is eliminated by the present invention.

(44) Turning now to FIG. **10**, an exploded view of an in-use embodiment is shown. Specifically, drainpipe **90**, such as a plumber trap commonly employed underneath a sink or washbasin **89**, mechanically couples to pipe connector apparatus **10** via straight section **91** that is disposed between output **95** and adjacent wall **97**. In particular, connector **10** is installed in the position between longitudinally separated sections A, B of drainpipe **90**; for example, a narrow portion of drainpipe **90** can be removed, thereby leaving exposed end edges **96** and **98**, respectively. Next, drainpipe locking rings **92** and **102**, and their respective retainer rings **94** and **100**, are placed on the separated drainpipe sections A, B. From there, cut end edges **96**, **98** are inserted into their corresponding first pipe opening **21** and section pipe opening **23** of main pipe member **12**. Once inserted into central passage **18**, locking rings **92**, **102** are rotated to mechanically engage male-threaded sections **28**, **30** in a tightened, sealed arrangement.

(45) Additionally, quick-release elbow connector **104** having opposed inlet **110** and output **108** ends

is coupled to exhaust hose **106** of medical gas scavenger device **112**. Elbow output **108** is then releasably coupled to riser **16**. Thus, any gas, or a mixture of gas, being exhausted through hose **106** by gas scavenger device **112** (as indicated by the arrow) flows through connector **10**, into central passage **18**, and out through second pipe opening **23** of main pipe member **12**, into exhaust riser **114** located behind wall **97** (see FIG. 1), and ultimately into the waste-collecting system of the building structure.

(46) Since the gas flowing through connector **10** nominally will be at atmospheric pressure, there is no tendency of the gas to disturb, or remove, or otherwise affect in any way the water column in drainpipe **90**. Specifically, the water column prevents any gas from the building's waste collection system from back-flowing through the sink drain into the room where the sink is located. And, by the same token, in accordance with the present invention, one-way valve assembly **72** in connector **10** prevents the backflow of any such gas through riser **16**.

(47) Turning to FIG. **11**, an alternative assembly is shown, including reversing elbow **120** that replaces drainpipe **90**. Reversing elbow **120** includes end edge **126**, second locking ring **122**, the corresponding retainer ring **124**, and shortened straight section **121** of drainpipe **120** adjacent to the depicted wall. When reversing elbow **120** is employed as depicted in FIG. **11**, only one cut (i.e., a single cut) needs be made to drainpipe straight section **121**. Once the cut is made, end edge **128** is inserted into section pipe opening **21** of main pipe member **12**. Next, lock ring **132** and retainer ring **130** are disposed on drainpipe **90**. Next, end edge **126** of reversing elbow **120** is disposed within first pipe opening **23** of in-line pipe connector apparatus **10**. Locking rings **122**, **123** are then tightened onto threaded end sections **28**, **30** of main pipe member **12**. In such embodiments, such installation of pipe connector **10** is suitable where washbasin **89** is located near an adjoining wall, where space underneath washbasin **89** is limited or a minimum amount of cutting (i.e., removing) of a portion of drainpipe **90** is desired.

(48) The components of the present invention can be made from inexpensive, durable molded plastic materials. Fabrication of the pipe connector apparatus **10** of the present invention is relatively easy and therefore inexpensive. In an embodiment, valve assembly **72** can be joined to valve housing **14**. By inserting spindle **76** into central opening **54** of hub **52**, valve housing **14** can be joined as by suitable gluing to support boss **32** of main pipe member **12**. Finally, riser **16** can be joined in place at the bottom of recess **46** (again as by suitable gluing).

(49) From the foregoing, it will be apparent that the pipe connector apparatus **10** of the present invention is suitable for advantageous use in a wide variety of applications. One such application, mentioned by way of example, and merely as illustrative and not to be construed as limiting, is to provide a convenient, effective, and safe means for removing the exhaled waste gas of a patient breathing a mixture of nitrous oxide and oxygen during a medical procedure being performed in a treatment room via an analgesia system, such as that commercially distributed by Sedation Systems LLC, St. Petersburg, FL, under the Nitrouseal® trademark. An embodiment of the Nitrouseal® analgesia system is disclosed in U.S. Pat. No. 8,826,905 B2, entitled "Respiratory Face Mask and Breathing Circuit Assembly," which is hereby incorporated by reference in its entirety. By further way of example, in the aforementioned Nitrouseal® analgesia system, a Miniscav® portable waste gas evacuation pump manufactured by RA Medical Svcs., Ltd., Steeton, England may be used as the "scavenger" or "vacuum" to remove expired waste gas from the treatment area where the equipment is being used.

(50) Piston-Valve Pipe Member

(51) FIGS. **12-16** depict pipe connector apparatus **10**, having main pipe member **12**, valve housing **14**, and riser **16**, extending upwardly away from valve housing portion **14**, according to embodiments of the present disclosure. As shown in FIGS. **12-13**, main pipe member **12** may be cylindrically shaped, defined by a hollow bore or central passage **18** extending between first peripheral end edge **20** and second peripheral edge **22**, such that central passage **18** may be formed throughout main pipe member **12**. In embodiments, first peripheral end edge **20** may define first

opening **21**, and second peripheral end edge **22** may define second opening **23**, such that central passage **18** may be formed throughout main pipe member **12**.

(52) Additionally, As depicted in FIGS. **12-13**, main pipe member **12** includes continuous interior wall surface **24** that may be substantially cylindrical and smooth, such that interior wall surface **24** is free from projections or unevenness. In contrast, exterior surface **26** of main pipe member **12** includes first and second spiral fastening sections **28, 30**, which may be male-threaded fastening sections, as shown in FIG. **13**. As previously stated, first and second spiral male-threaded fastening sections **28, 30** mechanically couple with a corresponding complementary spiral female-threaded sections of conventional drainpipe locking rings or collars. In an embodiment, fastening sections **28, 30** may be any fastening means, such that main pipe member **12** is mechanically coupled with a corresponding complementary fastening section of conventional drainpipe locking rings or collars.

(53) Piston-Valve Housing

(54) FIG. **14** depicts, in some embodiments, valve assembly **72** regulating the fluid flow between riser **16** and central passage **18** of main pipe member **12**. In some embodiments, the regulation of the fluid flow may be achieved by transitioning rigid (or stopper) portion **138** between a closed position and an open position. In some embodiments, flange gasket **136** is provided on rigid portion **138** and is spaced apart from biasing component **134**. Flange gasket **136** is sized to enable rigid portion **138** to abut and press-fit into floor member **50**. By doing so, valve assembly **72** is locked in place and supported by floor member **50** in the closed position, as shown in FIG. **14**.

(55) As shown in FIG. **14**, in some embodiments, valve assembly **72** may be biased toward the closed position, such that rigid portion **138** may abut floor member **50**. Accordingly, when valve assembly **72** is in the closed position, the nature of flange gasket **136** may completely seal off passages **58A, 58B**, and **58C** as shown in FIG. **15**, from being in communication with cavity **64**. Thus, the backflow of any fluid back into riser **16** may be prevented.

(56) As shown in FIGS. **14-16**, in some embodiments, when rigid portion **138** is acted upon by the force of a fluid (such as flowing gas from a medical gas scavenging device) traveling through central passage **70** of riser **16**, the force may act upon biasing component **134**, whereby biasing component **134** may be compressed and rigid portion **138** translates opposite the floor member **50** and assumes the open position. As shown in FIGS. **14-15**, at least one rib **56** extends radially between bottom end edge **68** forming passages **58A, 58B**, and **58C** (collectively referred to as passages **58**) As depicted in FIGS. **14-15**, in such open positions, any gas or other fluid flowing through riser **16** may continue to flow as indicated by arrow **80** through passages **58**. In some embodiments, the fluid then may flow into cavity **64** and through passageway **37**. In some embodiments, the fluid may exit into central passage **18** of main pipe member **12**, where the fluid can be safely discharged into the building's waste-collection system.

(57) Thus, in some embodiments, by providing valve assembly **72** in accordance with the present invention, the pipe connector apparatus **10** of FIGS. **13-16** may be capable only of uni-directional exhaust flow. Accordingly, any backflow of an undesirable fluid (e.g., sewer gas) into the treatment room or area that may emanate from the building's waste-collection system to which pipe connector apparatus **10** is connected may be eliminated by the present invention.

(58) Referring now to FIG. **17**, in conjunction with FIGS. **1-16**, an exemplary process flow diagram is provided, depicting a method for the removal of a medical waste gas from within a treatment room. The steps delineated in the exemplary process flow diagram of FIG. **17** are merely exemplary of a preferred order for the removal of a medical waste gas from a treatment room. The steps may be carried out in another order, with or without additional steps including therein. Additionally, the steps may be carried out with an alternative embodiment of pipe connection apparatus **10**, as contemplated in the above description.

(59) The method for the removal (or evacuation) of a medical gas from a treatment room of a building begins at step **200**, during which a drainpipe disposed between a washbasin and a wall of the treatment room is identified. At step **210**, a cutting device is placed onto the drainpipe. The

cutting device being configured to cut the drainpipe into a first section and a second section. At step **220**, the cutting device is operated, such that the drainpipe is cut into the first section and the second section. Next, the pipe connector apparatus is installed between the first section and the second section at step **230**. Specifically, at step **240**, a second trap end of a plumbing trap is coupled to a first member end of the pipe connector. The plumbing trap includes a first trap end opposite a second trap end, where the first trap end is coupled to a washbasin. The pipe connection apparatus includes the components discussed above. The method then proceeds to step **250**, in which the second member end of the pipe connector is coupled to a drainpipe. The drainpipe being in fluid communication with a waste collection system of the building. In step **260**, an exhaust hose in fluid communication is coupled with a medical gas scavenging device to the distal end of the valve housing. In step **270**, an amount of the medical waste gas emitted by the medical gas scavenging device is transferred, via the exhaust hose, into the pipe connector. Next, at step **280**, the amount of the medical waste gas is exhausted, via the drainpipe, into the waste collection system of the building.

(60) The advantages set forth above, and those made apparent from the foregoing description, are efficiently attained. Since certain changes may be made in the above construction without departing from the scope of the invention, it is intended that all matters contained in the foregoing description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

(61) It is also to be understood that the following claims are intended to cover all of the generic and specific features of the invention herein described and all statements of the scope of the invention that, as a matter of language, might be said to fall therebetween.

Claims

1. A pipe-connector assembly system for exhausting a waste gas from a building, the system comprising: a pipe connector apparatus including a main pipe member extending between a first member end and a second member end; an exhaust hose, the exhaust hose including: a first end fluidically coupled to a gas scavenger device, wherein the gas scavenger device is configured to capture exhaust gas resulting from operation of a nitrous oxide mixing analgesia system; a second end fluidically coupled to the pipe connector apparatus; a drainpipe having a first end coupled to the second member end of the pipe connector apparatus and a second end coupled to a waste collection system of the building; wherein the waste gas flows from the exhaust hose through the pipe connector apparatus and into the waste collection system of the building, such that the waste gas is safely evacuated from the building.
2. The pipe-connector assembly system of claim 1, further comprising: a quick-release valve having an inlet, an output, and a passage extending between the inlet and the output; the inlet configured to be removably coupled to the exhaust hose; and the output coupled to the pipe connector apparatus between the first member end and the second member end thereof; wherein the amount of the waste gas flows from the exhaust hose through the quick-release valve to the pipe connector apparatus and is exhausted into the waste collection system of the building.
3. The pipe-connector assembly system of claim 1, further comprising: an intermediary riser disposed between the pipe connector apparatus and the exhaust hose, the intermediary riser including: a body defining a bore extending between a first riser end fluidically coupled to the pipe connector apparatus and a second riser end fluidically coupled to the exhaust hose.
4. The pipe-connector assembly system of claim 1, further comprising: a plumbing trap, having a first trap end and a second trap end, wherein the first trap end is connected to a washbasin disposed within the interior environment, and wherein the second trap end is connected to the first member end of the pipe connector apparatus.
5. The pipe-connector assembly system of claim 4, further comprising: a first attachment

mechanism disposed at the first member end and configured to threadedly engage with a second attachment mechanism disposed at the second trap end of the plumbing trap; a third attachment mechanism disposed at the second member end and configured to threadedly engage with the first end of the drainpipe; wherein the pipe connector apparatus is secured between the plumbing trap and the drainpipe.

6. The pipe-connector assembly system of claim 1, further comprising: a one-way valve disposed between and in fluidic communication with each of the pipe connector apparatus and the exhaust hose, wherein the one-way valve is configured to prevent a backflow of the amount of the waste gas from the waste collection system back into the building.

7. The pipe-connector assembly system of claim 6, further comprising: a valve housing including a proximal end and an opposing distal end, the proximal end connected to the pipe connector apparatus between the first member end and the second member end thereof, and the distal end connected to the exhaust hose, wherein the valve resides within the valve housing between the proximal end and the distal end thereof.

8. The pipe-connector assembly system of claim 7, wherein the valve further comprises: a closed position and an open position, wherein the open position permits the amount of the waste gas to flow through the valve housing from an internal environment of the building to the waste collection system of the building and the closed position prevents the backflow of the amount of the waste gas from the waste collection system back into the internal environment; a biasing component applying a biasing force on a rigid portion towards the closed position; whereby the rigid portion is translatable between the closed position and the open position when an external force, emanating through the exhaust hose, overcomes the biasing force.

9. The pipe-connector assembly system of claim 1, wherein the waste gas at least partially includes nitrous oxide, oxygen, or a combination of nitrous oxide and oxygen.

10. A pipe-connector assembly system for exhausting a medical waste gas from a building, the system comprising: a plumbing trap having a first trap end and a second trap end, wherein the first trap end is connected to a washbasin disposed within an interior environment of the building; a pipe connector apparatus including a main pipe member extending between a first member end and a second member end, the first member end coupled to the plumbing trap; a medical gas scavenger device fluidically coupled to the pipe connector apparatus, the medical gas scavenger device configured to capture an amount of the medical waste gas resulting from operation of a nitrous oxide mixing analgesia system; and a drainpipe having a first end coupled to the second member end of the pipe connector apparatus and a second end coupled to a waste collection system of the building; wherein the amount of the medical waste gas flows from the medical gas scavenger device through the pipe connector apparatus and into the waste collection system of the building, such that the amount of the medical waste gas is safely evacuated from the interior environment of the building.

11. The pipe-connector assembly system of claim 10, further comprising: a quick-release valve having an inlet, an output, and a passage extending between the inlet and the output; the inlet configured to be removably coupled to an exhaust hose of the medical gas scavenger device; and the output coupled to the pipe connector apparatus between the first member end and the second member end thereof; wherein the amount of the medical waste gas flows from the medical gas scavenger device through the quick-release valve to the pipe connector apparatus and exhausted into the waste collection system of the building.

12. The pipe-connector assembly system of claim 10, further comprising: an intermediary riser disposed between the pipe connector apparatus and the medical gas scavenger device, the intermediate riser including: a body defining a bore extending between a first riser end fluidically coupled to the pipe connector apparatus and a second riser end fluidically coupled to an exhaust hose of the medical gas scavenger device.

13. The pipe-connector assembly system of claim 10, further comprising: a first attachment

mechanism disposed at the first member end and configured to threadedly engage with a second attachment mechanism disposed at the second trap end of the plumbing trap; a third attachment mechanism disposed at the second member end and configured to threadedly engage with the first end of the drainpipe; wherein the pipe connector apparatus is secured between the plumbing trap and the drainpipe.

14. The pipe-connector assembly system of claim 10, further comprising: a one-way valve disposed between and in fluidic communication with each of the pipe connector apparatus and the medical gas scavenger device, wherein the one-way valve is configured to prevent a backflow of the amount of the medical waste gas from the waste collection system back into the internal environment.

15. The pipe-connector assembly system of claim 14, further comprising: a valve housing including a proximal end and an opposing distal end, the proximal end connected to the pipe connector apparatus between the first member end and the second member end thereof, and the distal end connected to the medical gas scavenger device, wherein the valve resides within the valve housing between the proximal end and the distal end thereof.

16. The pipe-connector assembly system of claim 15, wherein the valve further comprises: a closed position and an open position, wherein the open position permits the amount of the medical waste gas to flow through the valve housing from the internal environment to the waste collection system of the building and the closed position prevents the backflow of the amount of the medical waste gas from the waste collection system back into the internal environment; a biasing component applying a biasing force on a rigid portion towards the closed position; whereby the rigid portion is translatable between the closed position and the open position when an external force, emanating from the medical gas scavenger device, overcomes the biasing force.

17. The pipe-connector assembly system of claim 15, wherein the valve housing further comprises: an interior wall extending between the proximal end and the distal end, the interior wall defining a channel therein; a support floor member disposed within the channel, the support floor member including a central hub defining a central valve-step support aperture and a rib extending radially between the central hub and the interior wall of the channel.

18. The pipe-connector assembly system of claim 10, wherein the waste gas at least partially includes nitrous oxide, oxygen, or a combination of nitrous oxide and oxygen.

19. A method of exhausting a medical waste gas scavenged by a medical gas scavenging device from a room in a building, the method comprising: coupling a second member end of a pipe connector apparatus to a drainpipe, the drainpipe fluidically coupled to a waste collection system of the building, the pipe connector apparatus including a first member end opposite the second member end and a main pipe member extending between the first member end and the second member end; coupling an exhaust hose to the medical gas scavenging device and to the pipe connector apparatus; operating a nitrous oxide mixing analgesia system; collecting, via the medical gas scavenging device, an amount of the medical waste gas resulting from the operation of the nitrous oxide mixing analgesia system; transferring, via the exhaust hose, the amount of the medical waste gas into the pipe connector apparatus; and exhausting, via the drainpipe, the amount of the medical waste gas into the waste collection system of the building.

20. The method of exhausting the medical waste gas of claim 18, further comprising the steps of: providing a quick release valve having an inlet, an output, and a passage extending between the inlet and the output; coupling the inlet to the exhaust hose of the medical gas scavenger device; coupling the output to the pipe connector apparatus; and transferring, via the quick-release valve, the medical waste gas from the medical gas scavenging device to the pipe connector apparatus.

21. The method of exhausting the medical waste gas of claim 18, further comprising the steps of: prior to the step of coupling the second member end of the pipe connector apparatus to the drainpipe, cutting the drainpipe into a first section and the second section and installing the pipe connector apparatus between the first section and the second section, such that the first member end

- of the pipe connector apparatus connects to the first section, and such that the second member end of the pipe connector apparatus connects to the second section.
22. The method of exhausting the medical waste gas of claim 18, wherein the waste gas at least partially includes nitrous oxide, oxygen, or a combination of nitrous oxide and oxygen.
23. A method of exhausting a medical waste gas scavenged by a medical gas scavenging device from a room in a building, the method comprising: coupling a scavenger-to-drainpipe connector to a drainpipe, wherein the drainpipe is fluidically coupled to a waste collection system of the building; establishing a fluidic channel between the medical gas scavenging device and the scavenger-to-drainpipe connector, wherein the fluidic channel is at least partially established by an exhaust hose; operating a nitrous oxide mixing analgesia system; collecting, via the medical gas scavenging device, an amount of the medical waste gas resulting from the operation of the nitrous oxide mixing analgesia system; transferring, via the fluidic channel, the amount of the medical waste gas into the drainpipe; and exhausting, via the drainpipe, the amount of the medical waste gas into the waste collection system of the building.
24. The method of claim 23, wherein the step of coupling the scavenger-to-drainpipe connector to the drainpipe includes sealingly mating the scavenger-to-drainpipe connector to the drainpipe.
25. The method of claim 23, wherein the scavenger-to-drainpipe connector is coupled to the drainpipe at a location downstream from a plumbing trap.
26. The method of claim 23, wherein the medical waste gas at least partially includes nitrous oxide, oxygen, or a combination of nitrous oxide and oxygen.
27. The method of claim 23, wherein the medical gas scavenging device is connected to a source of analgesia gas.
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